



## 1. Description

### 1.1. Project

|                 |                    |
|-----------------|--------------------|
| Project Name    | stm32_ble          |
| Board Name      | NUCLEO-F411RE      |
| Generated with: | STM32CubeMX 6.16.1 |
| Date            | 01/17/2026         |

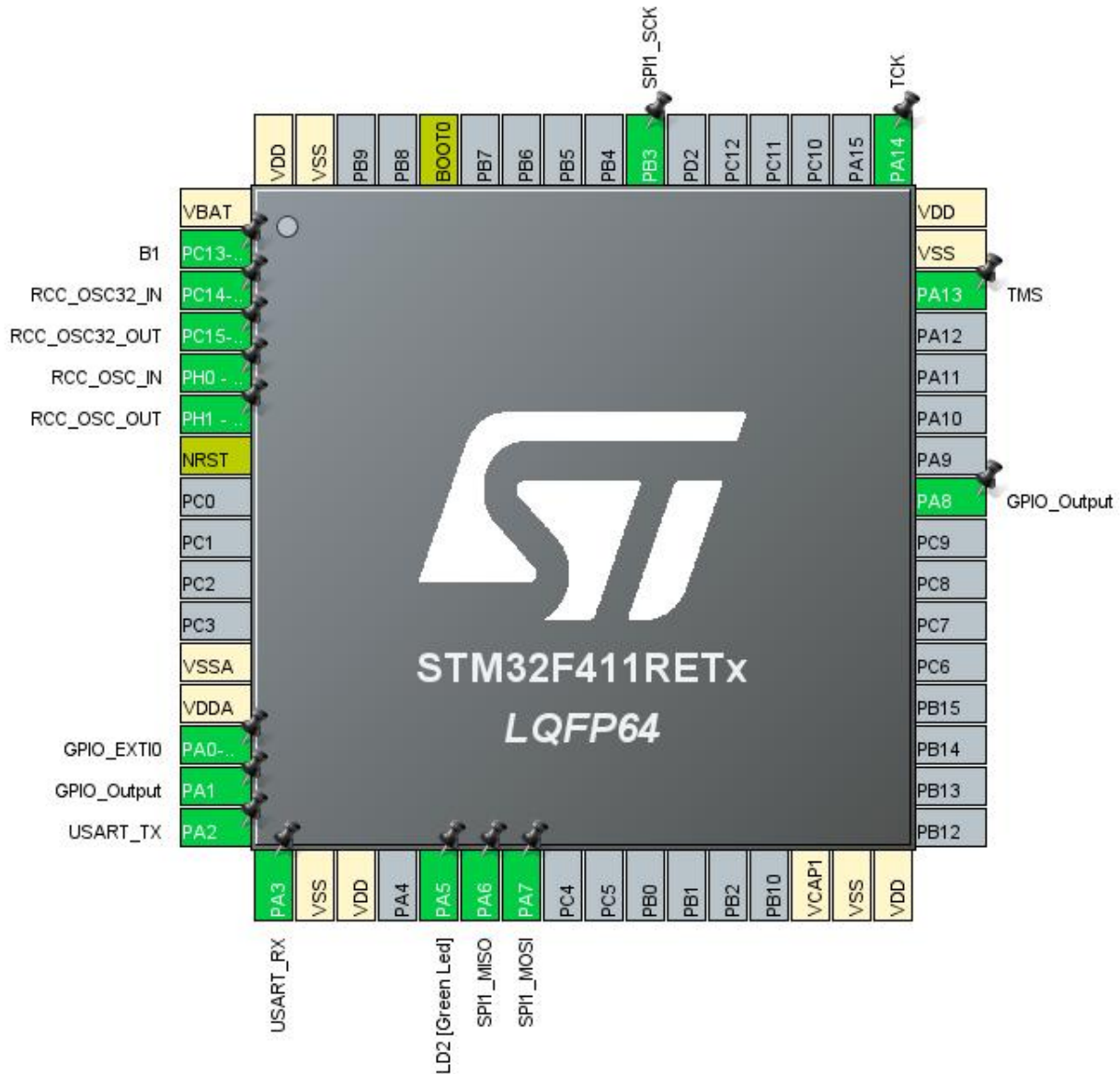
### 1.2. MCU

|                |               |
|----------------|---------------|
| MCU Series     | STM32F4       |
| MCU Line       | STM32F411     |
| MCU name       | STM32F411RETx |
| MCU Package    | LQFP64        |
| MCU Pin number | 64            |

### 1.3. Core(s) information

|         |               |
|---------|---------------|
| Core(s) | Arm Cortex-M4 |
|---------|---------------|

## 2. Pinout Configuration

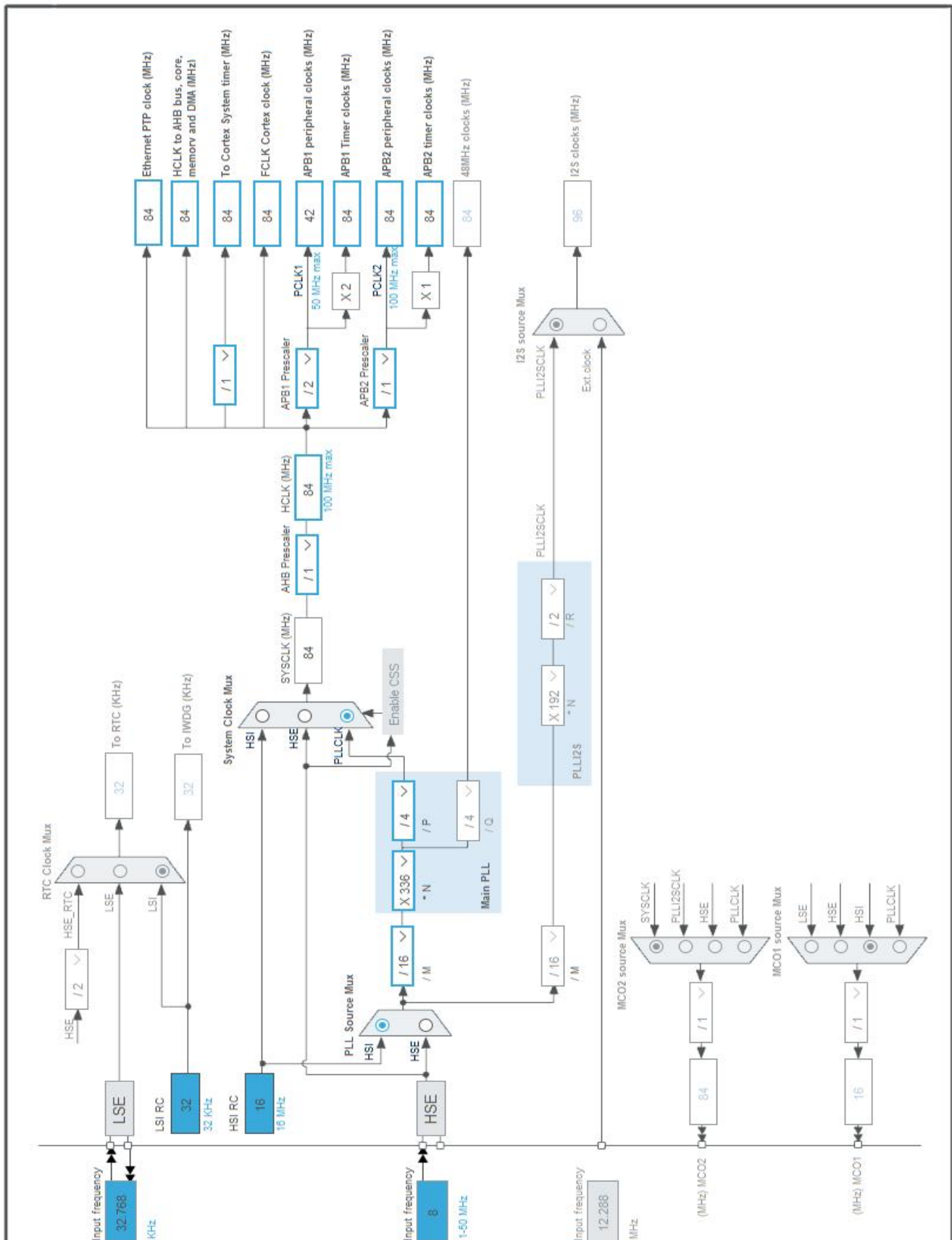


### 3. Pins Configuration

| Pin Number<br>LQFP64 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label           |
|----------------------|---------------------------------------|----------|--------------------------|-----------------|
| 1                    | VBAT                                  | Power    |                          |                 |
| 2                    | PC13-ANTI_TAMP                        | I/O      | GPIO_EXTI13              | B1              |
| 3                    | PC14-OSC32_IN                         | I/O      | RCC_OSC32_IN             |                 |
| 4                    | PC15-OSC32_OUT                        | I/O      | RCC_OSC32_OUT            |                 |
| 5                    | PH0 - OSC_IN                          | I/O      | RCC_OSC_IN               |                 |
| 6                    | PH1 - OSC_OUT                         | I/O      | RCC_OSC_OUT              |                 |
| 7                    | NRST                                  | Reset    |                          |                 |
| 12                   | VSSA                                  | Power    |                          |                 |
| 13                   | VDDA                                  | Power    |                          |                 |
| 14                   | PA0-WKUP                              | I/O      | GPIO_EXTI0               |                 |
| 15                   | PA1 *                                 | I/O      | GPIO_Output              |                 |
| 16                   | PA2                                   | I/O      | USART2_TX                | USART_TX        |
| 17                   | PA3                                   | I/O      | USART2_RX                | USART_RX        |
| 18                   | VSS                                   | Power    |                          |                 |
| 19                   | VDD                                   | Power    |                          |                 |
| 21                   | PA5 *                                 | I/O      | GPIO_Output              | LD2 [Green Led] |
| 22                   | PA6                                   | I/O      | SPI1_MISO                |                 |
| 23                   | PA7                                   | I/O      | SPI1_MOSI                |                 |
| 30                   | VCAP1                                 | Power    |                          |                 |
| 31                   | VSS                                   | Power    |                          |                 |
| 32                   | VDD                                   | Power    |                          |                 |
| 41                   | PA8 *                                 | I/O      | GPIO_Output              |                 |
| 46                   | PA13                                  | I/O      | SYS_JTMS-SWDIO           | TMS             |
| 47                   | VSS                                   | Power    |                          |                 |
| 48                   | VDD                                   | Power    |                          |                 |
| 49                   | PA14                                  | I/O      | SYS_JTCK-SWCLK           | TCK             |
| 55                   | PB3                                   | I/O      | SPI1_SCK                 |                 |
| 60                   | BOOT0                                 | Boot     |                          |                 |
| 63                   | VSS                                   | Power    |                          |                 |
| 64                   | VDD                                   | Power    |                          |                 |

\* The pin is affected with an I/O function

## 4. Clock Tree Configuration



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |               |
|-----------|---------------|
| Series    | STM32F4       |
| Line      | STM32F411     |
| MCU       | STM32F411RETx |
| Datasheet | DS10314 Rev6  |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 1.7 |

### 1.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

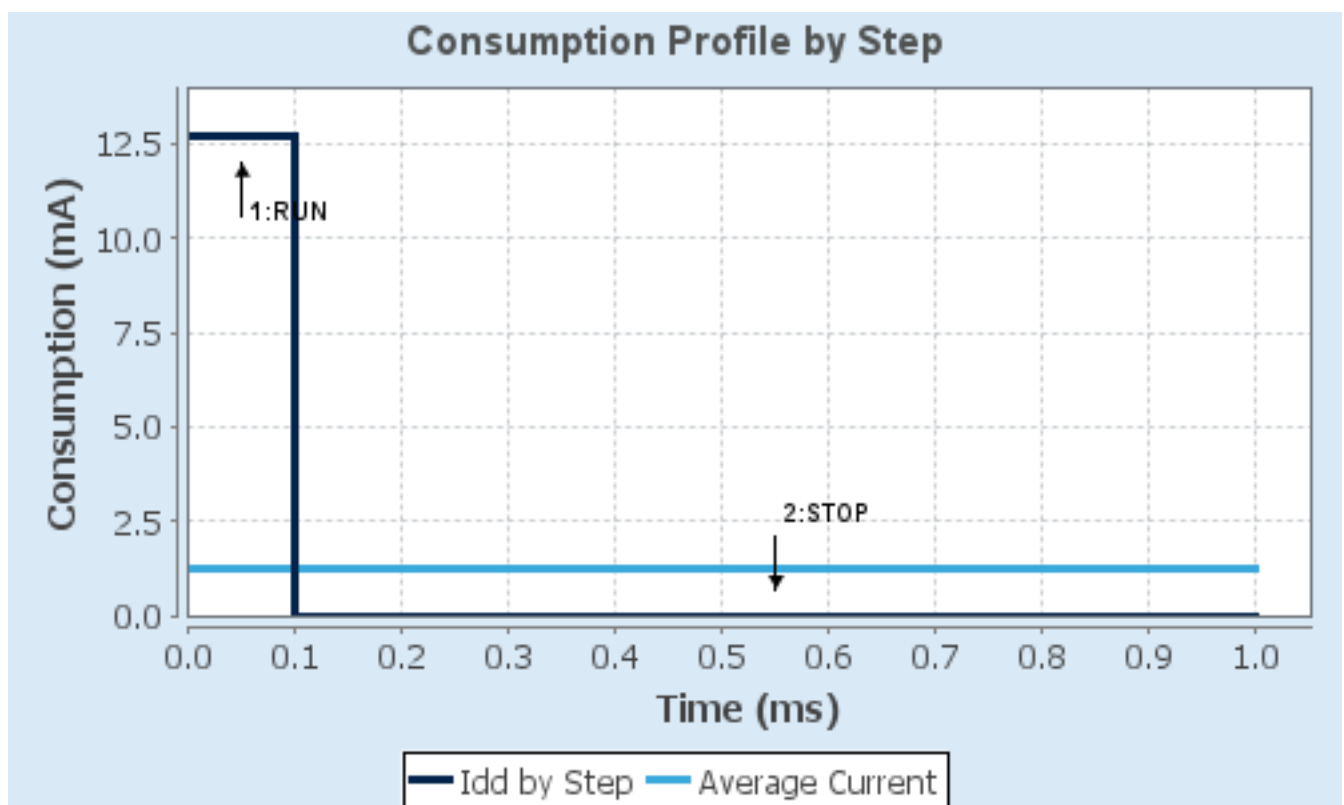
#### 1.4. Sequence

|                               |             |                             |
|-------------------------------|-------------|-----------------------------|
| <b>Step</b>                   | Step1       | Step2                       |
| <b>Mode</b>                   | RUN         | STOP                        |
| <b>Vdd</b>                    | 1.7         | 1.7                         |
| <b>Voltage Source</b>         | Battery     | Battery                     |
| <b>Range</b>                  | Scale1-High | No Scale                    |
| <b>Fetch Type</b>             | SRAM        | n/a                         |
| <b>CPU Frequency</b>          | 100 MHz     | 0 Hz                        |
| <b>Clock Configuration</b>    | HSE PLL     | Regulator_LPLV Flash-PwrDwn |
| <b>Clock Source Frequency</b> | 4 MHz       | 0 Hz                        |
| <b>Peripherals</b>            |             |                             |
| <b>Additional Cons.</b>       | 0 mA        | 0 mA                        |
| <b>Average Current</b>        | 12.7 mA     | 9 $\mu$ A                   |
| <b>Duration</b>               | 0.1 ms      | 0.9 ms                      |
| <b>DMIPS</b>                  | 125.0       | 0.0                         |
| <b>Ta Max</b>                 | 103.99      | 105                         |
| <b>Category</b>               | In DS Table | In DS Table                 |

#### 1.5. Results

|               |                            |                 |             |
|---------------|----------------------------|-----------------|-------------|
| Sequence Time | 1 ms                       | Average Current | 1.28 mA     |
| Battery Life  | 3 months, 19 days, 6 hours | Average DMIPS   | 125.0 DMIPS |

#### 1.6. Chart





## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                                       |
|-----------------------------------|---------------------------------------------|
| Project Name                      | stm32_ble                                   |
| Project Folder                    | D:\Workspace\STM32_BLE\BLE_Device\stm32_ble |
| Toolchain / IDE                   | STM32CubeIDE                                |
| Firmware Package Name and Version | STM32Cube FW_F4 V1.28.3                     |
| Application Structure             | Advanced                                    |
| Generate Under Root               | Yes                                         |
| Do not generate the main()        | No                                          |
| Minimum Heap Size                 | 0x200                                       |
| Minimum Stack Size                | 0x400                                       |

### 2.2. Code Generation Settings

| Name                                                            | Value                                 |
|-----------------------------------------------------------------|---------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                    |
| Backup previously generated files when re-generating            | No                                    |
| Keep User Code when re-generating                               | Yes                                   |
| Delete previously generated files when not re-generated         | Yes                                   |
| Set all free pins as analog (to optimize the power consumption) | No                                    |
| Enable Full Assert                                              | No                                    |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name       | Peripheral Instance Name |
|------|---------------------|--------------------------|
| 1    | SystemClock_Config  | RCC                      |
| 2    | MX_GPIO_Init        | GPIO                     |
| 3    | MX_USART2_UART_Init | USART2                   |

## 3. Peripherals and Middlewares Configuration

### 3.1. RCC

**High Speed Clock (HSE): BYPASS Clock Source**

**Low Speed Clock (LSE) : Crystal/Ceramic Resonator**

#### 3.1.1. Parameter Settings:

##### **System Parameters:**

|                   |                    |
|-------------------|--------------------|
| VDD voltage (V)   | 3.3                |
| Instruction Cache | Enabled            |
| Prefetch Buffer   | Enabled            |
| Data Cache        | Enabled            |
| Flash Latency(WS) | 2 WS (3 CPU cycle) |

##### **RCC Parameters:**

|                                |          |
|--------------------------------|----------|
| HSI Calibration Value          | 16       |
| TIM Prescaler Selection        | Disabled |
| HSE Startup Timeout Value (ms) | 100      |
| LSE Startup Timeout Value (ms) | 5000     |

##### **Power Parameters:**

|                               |                                 |
|-------------------------------|---------------------------------|
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 1 |
|-------------------------------|---------------------------------|

### 3.2. SPI1

**Mode: Full-Duplex Master**

#### 3.2.1. Parameter Settings:

##### **Basic Parameters:**

|              |           |
|--------------|-----------|
| Frame Format | Motorola  |
| Data Size    | 8 Bits    |
| First Bit    | MSB First |

##### **Clock Parameters:**

|                           |                         |
|---------------------------|-------------------------|
| Prescaler (for Baud Rate) | <b>64 *</b>             |
| Baud Rate                 | <b>1.3125 MBits/s *</b> |
| Clock Polarity (CPOL)     | Low                     |
| Clock Phase (CPHA)        | <b>2 Edge *</b>         |

##### **Advanced Parameters:**

|                 |          |
|-----------------|----------|
| CRC Calculation | Disabled |
| NSS Signal Type | Software |

### 3.3. SYS

**Debug: Serial Wire**

**Timebase Source: SysTick**

### 3.4. USART2

**Mode: Asynchronous**

#### 3.4.1. Parameter Settings:

##### **Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

##### **Advanced Parameters:**

|                |                      |
|----------------|----------------------|
| Data Direction | Receive and Transmit |
| Over Sampling  | 16 Samples           |

### 3.5. STMicroelectronics.X-CUBE-BLE2.3.3.0

**mode: WirelessJjBlueNRGAa2**

#### 3.5.1. Parameter Settings:

##### **Log & Debug:**

|                  |                                       |
|------------------|---------------------------------------|
| BLE2_DEBUG       | 0: No debug message                   |
| PRINT_CSV_FORMAT | 0: .csv format message print disabled |
| BLUENRG2_DEBUG   | 0: No debug message                   |

##### **Basic Parameters:**

|                         |     |
|-------------------------|-----|
| HCI_READ_PACKET_SIZE    | 128 |
| HCI_MAX_PAYLOAD_SIZE    | 128 |
| HCI_READ_PACKET_NUM_MAX | 10  |

##### **Connection Parameters (for expert users):**

|                                      |       |
|--------------------------------------|-------|
| Scan Interval (SCAN_P)               | 16384 |
| Scan Window (SCAN_L)                 | 16384 |
| Supervision Timeout (SUPERV_TIMEOUT) | 60    |
| Min Connection Period (CONN_P1)      | 40    |
| Max Connection Period (CONN_P2)      | 40    |
| Min Connection Length (CONN_L1)      | 2000  |

|                                                  |                                              |
|--------------------------------------------------|----------------------------------------------|
| Max Connection Length (CONN_L2)                  | 2000                                         |
| Advertising Type (ADV_DATA_TYPE)                 | Connectable Undirected Advertising (ADV_IND) |
| Min Advertising Interval (ADV_INTERV_MIN)        | 2048                                         |
| Max Advertising Interval (ADV_INTERV_MAX)        | 4096                                         |
| Min Connection Event Interval (L2CAP_INTERV_MIN) | 9                                            |
| Max Connection Event Interval (L2CAP_INTERV_MAX) | 20                                           |
| Timeout Multiplier (L2CAP_TIMEOUT_MULTIPLIER)    | 600                                          |

### 3.5.2. Platform Settings:

|               |          |
|---------------|----------|
| Exti Line     | PA0-WKUP |
| BUS IO driver | SPI1     |
| CS Line       | PA1      |
| Reset Line    | PA8      |

**\* User modified value**

## 4. System Configuration

### 4.1. GPIO configuration

| IP     | Pin            | Signal         | GPIO mode                                                  | GPIO pull/up pull down      | Max Speed   | User Label      |
|--------|----------------|----------------|------------------------------------------------------------|-----------------------------|-------------|-----------------|
| RCC    | PC14-OSC32_IN  | RCC_OSC32_IN   | n/a                                                        | n/a                         | n/a         |                 |
|        | PC15-OSC32_OUT | RCC_OSC32_OUT  | n/a                                                        | n/a                         | n/a         |                 |
|        | PH0 - OSC_IN   | RCC_OSC_IN     | n/a                                                        | n/a                         | n/a         |                 |
|        | PH1 - OSC_OUT  | RCC_OSC_OUT    | n/a                                                        | n/a                         | n/a         |                 |
| SPI1   | PA6            | SPI1_MISO      | Alternate Function Push Pull                               | No pull-up and no pull-down | Very High * |                 |
|        | PA7            | SPI1_MOSI      | Alternate Function Push Pull                               | No pull-up and no pull-down | Very High * |                 |
|        | PB3            | SPI1_SCK       | Alternate Function Push Pull                               | No pull-up and no pull-down | Very High * |                 |
| SYS    | PA13           | SYS_JTMS-SWDIO | n/a                                                        | n/a                         | n/a         | TMS             |
|        | PA14           | SYS_JTCK-SWCLK | n/a                                                        | n/a                         | n/a         | TCK             |
| USART2 | PA2            | USART2_TX      | Alternate Function Push Pull                               | No pull-up and no pull-down | Very High * | USART_TX        |
|        | PA3            | USART2_RX      | Alternate Function Push Pull                               | No pull-up and no pull-down | Very High * | USART_RX        |
| GPIO   | PC13-ANTI_TAMP | GPIO_EXTI13    | External Interrupt Mode with Rising edge trigger detection | No pull-up and no pull-down | n/a         | B1              |
|        | PA0-WKUP       | GPIO_EXTI0     | External Interrupt Mode with Rising edge trigger detection | No pull-up and no pull-down | n/a         |                 |
|        | PA1            | GPIO_Output    | Output Push Pull                                           | No pull-up and no pull-down | Low         |                 |
|        | PA5            | GPIO_Output    | Output Push Pull                                           | No pull-up and no pull-down | Low         | LD2 [Green Led] |
|        | PA8            | GPIO_Output    | Output Push Pull                                           | No pull-up and no pull-down | Low         |                 |

### 4.2. DMA configuration

nothing configured in DMA service

### 4.3. NVIC configuration

#### 4.3.1. NVIC

| Interrupt Table                         | Enable | Preenmption Priority | SubPriority |
|-----------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                  | true   | 0                    | 0           |
| Hard fault interrupt                    | true   | 0                    | 0           |
| Memory management fault                 | true   | 0                    | 0           |
| Pre-fetch fault, memory access fault    | true   | 0                    | 0           |
| Undefined instruction or illegal state  | true   | 0                    | 0           |
| System service call via SWI instruction | true   | 0                    | 0           |
| Debug monitor                           | true   | 0                    | 0           |
| Pendable request for system service     | true   | 0                    | 0           |
| System tick timer                       | true   | 0                    | 0           |
| EXTI line0 interrupt                    | true   | 0                    | 0           |
| EXTI line[15:10] interrupts             | true   | 0                    | 0           |
| PVD interrupt through EXTI line 16      | unused |                      |             |
| Flash global interrupt                  | unused |                      |             |
| RCC global interrupt                    | unused |                      |             |
| SPI1 global interrupt                   | unused |                      |             |
| USART2 global interrupt                 | unused |                      |             |
| FPU global interrupt                    | unused |                      |             |

#### 4.3.2. NVIC Code generation

| Enabled interrupt Table                 | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|-----------------------------------------|-----------------------------------|----------------------|------------------|
| Non maskable interrupt                  | false                             | true                 | false            |
| Hard fault interrupt                    | false                             | true                 | false            |
| Memory management fault                 | false                             | true                 | false            |
| Pre-fetch fault, memory access fault    | false                             | true                 | false            |
| Undefined instruction or illegal state  | false                             | true                 | false            |
| System service call via SWI instruction | false                             | true                 | false            |
| Debug monitor                           | false                             | true                 | false            |
| Pendable request for system service     | false                             | true                 | false            |
| System tick timer                       | false                             | true                 | true             |
| EXTI line0 interrupt                    | false                             | true                 | true             |
| EXTI line[15:10] interrupts             | false                             | true                 | true             |

\* User modified value

## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

## 6. Software Pack Report

### 6.1. Software Pack selected

| Vendor             | Name        | Version | Component                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|-------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STMicroelectronics | X-CUBE-BLE2 | 3.3.0   | Class : Wireless<br>Group :<br>BlueNRG-2<br>SubGroup :<br>Controller<br>Version : 3.3.0<br>Class : Wireless<br>Group :<br>BlueNRG-2<br>SubGroup :<br>HCI_TL<br>Variant : Basic<br>Version : 3.3.0<br>Class : Wireless<br>Group :<br>BlueNRG-2<br>SubGroup :<br>HCI_TL_INTERFACE<br>Variant :<br>UserBoard<br>Version : 3.3.0<br>Class : Wireless<br>Group :<br>BlueNRG-2<br>SubGroup : Utils<br>Version : 3.3.0 |



## 7. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BSDL files              | <a href="https://www.st.com/resource/en/bsdl_model/stm32f411_bsd1.zip">https://www.st.com/resource/en/bsdl_model/stm32f411_bsd1.zip</a>                                                                                                                                                                                                 |
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32f411_ibis.zip">https://www.st.com/resource/en/ibis_model/stm32f411_ibis.zip</a>                                                                                                                                                                                                 |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32f4-svd.zip">https://www.st.com/resource/en/svd/stm32f4-svd.zip</a>                                                                                                                                                                                                                     |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf</a>                                                                                                                             |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf">https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf</a>                                                                                                                                                     |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf">https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf</a>                                                                                                                               |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf</a>                                                                                                                               |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf</a>                                                                                                                             |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf">https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf</a>                                                                                                                                     |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf">https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf</a>                                                                                                                                   |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32f4x1.pdf">https://www.st.com/resource/en/flyer/flstm32f4x1.pdf</a>                                                                                                                                                                                                                 |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32nucleo.pdf">https://www.st.com/resource/en/flyer/flstm32nucleo.pdf</a>                                                                                                                                                                                                             |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32trust.pdf">https://www.st.com/resource/en/flyer/flstm32trust.pdf</a>                                                                                                                                                                                                               |
| Product Certifications  | <a href="https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf">https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf</a>                                                                                                                                                     |
| Security Bulletin       | <a href="https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf</a> |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf</a>                                                               |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an2945-stm8s-and-">https://www.st.com/resource/en/application_note/an2945-stm8s-and-</a>                                                                                                                                                                                       |

stm32-mcus-a-consistent-832bit-product-line-for-painless-migration-stmicroelectronics.pdf

Application Notes [https://www.st.com/resource/en/application\\_note/an3070-managing-the-driver-enable-signal-for-rs485-and-iolink-communications-with-the-stm32s-usart-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3070-managing-the-driver-enable-signal-for-rs485-and-iolink-communications-with-the-stm32s-usart-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an3364-migration-and-compatibility-guidelines-for-stm32-microcontroller-applications-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3364-migration-and-compatibility-guidelines-for-stm32-microcontroller-applications-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an3997-audio-playback-and-recording-using-the-stm32f4discovery-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3997-audio-playback-and-recording-using-the-stm32f4discovery-stmicroelectronics.pdf)

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| for related Tools<br>& Software                      | <a href="#">with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf</a>                                                                                                                                                                                                                                                                                          |
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| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf</a>                                                   |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5464-position-control-of-a-three-phase-permanent-magnet-motor-using-xcubemcsdk-or-xcubemcsdkful-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5464-position-control-of-a-three-phase-permanent-magnet-motor-using-xcubemcsdk-or-xcubemcsdkful-stmicroelectronics.pdf</a>                           |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dual-core-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dual-core-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf</a>                             |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5698-adapting-the-xcubestl-functional-safety-package-for-stm32-iec-61508-compliant-to-other-safety-standards-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5698-adapting-the-xcubestl-functional-safety-package-for-stm32-iec-61508-compliant-to-other-safety-standards-stmicroelectronics.pdf</a> |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf</a>                                                                                                               |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an4502-stm32-smbus-pmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an4502-stm32-smbus-pmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf</a>                                                                                                             |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf</a>                                                                                                                                               |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf</a>                                                                                         |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an6179-how-to-integrate-the-stl-firmware-into-a-time-critical-user-application-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6179-how-to-integrate-the-stl-firmware-into-a-time-critical-user-application-stmicroelectronics.pdf</a>                                                                 |
| Application Notes<br>for related Tools               | <a href="https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf</a>                                                                                                   |



## & Software

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| Application Notes for related Tools & Software | <a href="https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf</a>                                                                                 |
| Design Notes & Tips                            | <a href="https://www.st.com/resource/en/design_tip/dt0088-fir-filter-design-by-sampling-windowing-and-modulating-the-sinc-function-stmicroelectronics.pdf">https://www.st.com/resource/en/design_tip/dt0088-fir-filter-design-by-sampling-windowing-and-modulating-the-sinc-function-stmicroelectronics.pdf</a>                                               |
| Design Notes & Tips                            | <a href="https://www.st.com/resource/en/design_tip/dt0089-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-stmicroelectronics.pdf">https://www.st.com/resource/en/design_tip/dt0089-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-stmicroelectronics.pdf</a>               |
| Design Notes & Tips                            | <a href="https://www.st.com/resource/en/design_tip/dt0091-lattice-wave-digital-filter-design-and-automatic-c-code-generation-stmicroelectronics.pdf">https://www.st.com/resource/en/design_tip/dt0091-lattice-wave-digital-filter-design-and-automatic-c-code-generation-stmicroelectronics.pdf</a>                                                           |
| Design Notes & Tips                            | <a href="https://www.st.com/resource/en/design_tip/dt0092-lattice-wave-digital-filter-test-and-performance-verification-stmicroelectronics.pdf">https://www.st.com/resource/en/design_tip/dt0092-lattice-wave-digital-filter-test-and-performance-verification-stmicroelectronics.pdf</a>                                                                     |
| Device Option Lists                            | <a href="https://www.st.com/resource/en/device_option_list/opl_stm32f411_512k.zip">https://www.st.com/resource/en/device_option_list/opl_stm32f411_512k.zip</a>                                                                                                                                                                                               |
| Errata Sheets                                  | <a href="https://www.st.com/resource/en/errata_sheet/es0287-stm32f411xcxe-device-errata-stmicroelectronics.pdf">https://www.st.com/resource/en/errata_sheet/es0287-stm32f411xcxe-device-errata-stmicroelectronics.pdf</a>                                                                                                                                     |
| Datasheet                                      | <a href="https://www.st.com/resource/en/datasheet/dm00115249.pdf">https://www.st.com/resource/en/datasheet/dm00115249.pdf</a>                                                                                                                                                                                                                                 |
| Programming Manuals                            | <a href="https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf">https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf</a>                                                                                 |
| Reference Manuals                              | <a href="https://www.st.com/resource/en/reference_manual/rm0383-stm32f411xce-advanced-armbased-32bit-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/reference_manual/rm0383-stm32f411xce-advanced-armbased-32bit-mcus-stmicroelectronics.pdf</a>                                                                                                 |
| Technical Notes & Articles                     | <a href="https://www.st.com/resource/en/technical_note/tn0516-overview-of-the-stm32f0xf100xxf103xx-and-stm32f2xxf30xf4xx-mcus-pmsm-singledual-foc-sdk-v40-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn0516-overview-of-the-stm32f0xf100xxf103xx-and-stm32f2xxf30xf4xx-mcus-pmsm-singledual-foc-sdk-v40-stmicroelectronics.pdf</a> |
| Technical Notes & Articles                     | <a href="https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf</a>                                     |
| Technical Notes & Articles                     | <a href="https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf</a>                                         |
| Technical Notes                                | <a href="https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-">https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-</a>                                                                                                                                                                                                         |

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| & Articles                    | <a href="#">shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf</a>                                                                                                                                                                                                                                                     |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf</a>                       |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf</a>                         |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf</a> |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf</a>                   |
| User Manuals                  | <a href="https://www.st.com/resource/en/user_manual/um3461-stm32f4-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf">https://www.st.com/resource/en/user_manual/um3461-stm32f4-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf</a>                                                             |